Cross-Temporal Sinhala OCR: Page-Level Adaptation and Diachronic Analysis

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***Abstract*—Sinhala is a morphologically rich abugida spoken by roughly 16 million people in Sri Lanka, and currently, no real-world datasets are publicly available for page-level Sinhala OCR. All previous studies for assessing Sinhala OCR models have used artificially generated data. To bridge the gap, we introduce `sinhala-ocr-lk-acts-1010`, an annotated dataset of 1,010 page-level images and their transcriptions collected from Sri Lankan Legislative Acts published between 1981-1989 and 2000-2019, split into 707 training examples, 101 validation examples, and 202 testing examples. Three models based on deep learning-based visual language processing, namely DeepSeek-OCR V1, DeepSeek-OCR V2, and LightOnOCR-2-1B, are fine-tuned using QLoRA in 8 experiments conducted on consumer and cloud GPUs. LightOnOCR-2-1B is the top performer, achieving a CER of 1.05% across all test examples, outperforming state-of-the-art open-source OCR models such as Surya-OCR (8.84%) and commercially available OCR models such as Tesseract v5 (10.69%) and Google Document AI (2.06%). Our results suggest that LightOnOCR-2-1B outperforms other baselines on real-world OCR tasks and maintains consistent performance across all print periods, even when documents are severely degraded.**

***Index Terms*—Optical Character Recognition, Sinhala, Low-resource Languages, Diachronic Evaluation**

I. Introduction

Optical character recognition is a key enabling technology for digitising printed documents and making their content searchable and accessible at scale. For low-resource and complex-script languages, however, OCR accuracy remains significantly lower than for high-resource languages such as English [1], [2].

Sinhala is the primary official language in Sri Lanka. The script of the Sinhala language is an abugida, with many characters, ligatures, and similarities among characters, making it challenging for OCR engines. Existing research has steadily improved the development of OCR engines using Tesseract that can handle Sinhala characters and the development of parallel corpora from government PDFs [3], [4]. Recent benchmarking has even been conducted on both commercially available and open-source OCR engines to compare their performance in zero-shot learning on synthetic images [2], [5]. However, all existing evaluations are conducted on synthetic images or limited font sets, and no publicly available dataset of real printed Sinhala documents exists.

Page-level VLMs are compatible with such a challenge by processing the whole image of the page in one forward pass, thus bypassing errors associated with a step-by-step layout detection and segmentation process [6], [7]. Adaptation of multi-billion-parameter VLMs is made possible due to parameter-efficient fine-tuning based on the QLoRA technique [8], which makes adapting billion-parameter VLMs feasible on consumer hardware, as recently demonstrated for low-resource Indic scripts [9].

A further gap is *diachronicity*; none of the Sinhala OCR research studies have quantified the effect of temporal variance on accuracy scores, despite proof that recognition accuracy falls from 87% for contemporary book scans to 67% for newspaper articles from the 1980s [10], commercial engines also deteriorate with older media. [11].

In this paper, we address both gaps. First, we release `sinhala-ocr-lk-acts-1010`[1], a dataset of 1,010 hand-corrected page-level image-text pairs from Sri Lankan legislative acts (1981 -1989, 2000-2019), split into 707 train, 101 validation, and 202 test pairs, made publicly available on Hugging Face. Second, we fine-tune three VLMs DeepSeek-OCR V1 [12], DeepSeek-OCR V2 [13], and LightOnOCR-2-1B [14] across eight LoRa [15] QLoRA [8] experiments,achieving a best CER of 1.05%, surpassing all open-source baselines and Google Document AI (2.06%). Third, we conduct the first diachronic evaluation of page-level Sinhala OCR across three temporal periods (1981-1989, 2000-2009, 2010-2019) [11], [16], [17].

II. Related Work

A. OCR Architectures and Scene Text Systems

The development of early deep learning OCR led to sequence-based recognition becoming the paradigm. The CRNN model [18] utilised CNN-based feature extraction and bidirectional LSTMs, in conjunction with a CTC decoder, without segmentation. In contrast, attention-based rectification architectures such as ASTER [19] and RARE [20] achieved better text recognition on curved and distorted text. Following the introduction of the transformer architecture [21], which sparked a revolution in vision tasks by introducing the vision transformer [22], the TrOCR model has achieved recent state-of-the-art results in text recognition [6], which surpassed all previous models using an encoder-decoder approach without

[1] https://huggingface.co/datasets/avishadilhara/sinhala-ocr-lk-acts-1010

any external language model support. In the realm of detection, the EAST [23] and CRAFT [24] advanced single-stage text localisation for arbitrary-shaped text, while PhotoOCR [25] demonstrated robust recognition in uncontrolled real-world conditions.

B. Low-Resource, Multilingual, and Sinhala OCR

OCR for low-resource languages remains challenging due to data absence, script complexity, and limited pre-training coverage [1]. Training on authentic over synthetic data is crucial whenever possible [26], which is particularly applicable to our project on Sinhala. Kolavi [9] introduced Nayana, an approach that adapts VLMs for OCR in 10 low-resource Indic languages using the LoRA technique and synthetic data, cutting down CER to less than a third of the baseline model. Hegghammer [11] found that commercial engines are superior to Tesseract when dealing with noisy texts in non-English scripts. On the specific topic of Sinhala, Vasantharajan et al. [4] fine-tuned Tesseract on more than 20 legacy fonts and reduced the initial CER from 7.61% to 4.74%, creating a parallel dataset based on government documents, which is the closest work in terms of dataset type. Jayatilleke and de Silva [2] compared six engines' zero-shot performance on synthetic Sinhala-Tamil data and found that Surya achieved the lowest WER of 2.61% for Sinhala; however, only clean synthetic images were used for evaluation. Purushoth and Ambegoda [5] benchmarked several open-source document image analysis models for Sinhala, finding that Surya-OCR provided the most balanced and accurate performance over legacy models such as Tesseract. The study by Anuradha et al. [10] reported over 87% accuracy on modern book text but only 67% on late 19th-century newspapers (1870–1890), providing the first empirical evidence of diachronic degradation in Sinhala OCR.

C. Page-Level OCR, Datasets, and Diachronic Evaluation

The lack of ground-truth annotated document images has often held back research in OCR [27], [28]. While the `lk_legal_docs` dataset [29] contains 230,091 Sri Lankan government documents, it lacks ground-truth data at the page level, which this paper provides. Standard OCR pipelines often suffer from error cascading across consecutive stages [30], [31]; page-level VLMs circumvent this problem by processing an entire page in a single pass. KOSMOS-2.5 [7], [31] shows the capacity of multimodal LLMs to understand text-heavy document images as a whole. However, LightOnOCR-2-1B [14], and DeepSeek-OCR [12] have shown that it is possible to achieve similar results even in the case of higher resolution document images by employing efficient vision token embeddings enabled by QLoRA [8] and LoRa [15]. Diachronic evaluation of a model by tracing its performance across different print eras has been identified as essential for measuring its robustness to typographic changes and media decay [16], [17]. Despite the documented diachronic degradation in Sinhala [10], all prior Sinhala OCR evaluations have been synchronic. This work conducts The first controlled diachronic evaluation for Sinhala, spanning 1981–2019.

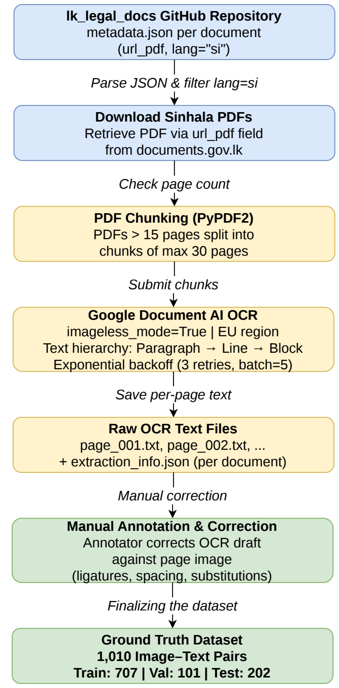

Fig. 1. Document processing pipeline for constructing the Sinhala government acts OCR dataset.

III. Dataset Preparation and Preprocessing

A. Source Documents

The dataset is sourced from the `lk_legal_docs` GitHub repository [29], a multilingual resource of Sri Lankan government documents. Each document folder contains a `metadata.json` file with fields including `doc_type`, `date_str`, `lang`, and `url_pdf` pointing to the PDF on `documents.gov.lk`. Only Sinhala-language (`lang: "si"`) documents were used; PDFs were downloaded programmatically via the `url_pdf` field. as illustrated in Fig. 1.

B. Document Processing Pipeline

PDFs were downloaded programmatically and split into 30-page chunks using PyPDF2 for documents exceeding 15 pages. Each chunk was submitted to Google Document AI [32] via the `google-cloud-document-ai` Python client with exponential backoff retry logic (3 retries, 60 s inter-batch delay); the returned per-page text was saved as UTF-8 `.txt` files and used as the manual annotation seed, as illustrated in Fig. 1.

C. Page Selection and Manual Annotation

Pages that had well-formed paragraphs in Sinhala were selected for use in our analysis, and other pages, such as forms and tables were not included for the sake of uniformity. Ground-truthing involved manually fixing errors made by Document AI [32], such as ligature recognition, character substitutions, inconsistent spacing, and erroneous line breaks in the tex t.Documents from the 1980s needed the most

TABLE I
SINHALA GOVERNMENT ACTS OCR DATASET STATISTICS.

Attribute	Value
Document type	Government acts (Sinhala)
Year range	1981–1989, 2000–2019
Total annotated pages	1,010
1981–1989	410
2000–2009	300
2010–2019	300
Training pairs	707 (70%)
Validation pairs	101 (10%)
Test pairs	202 (20%)

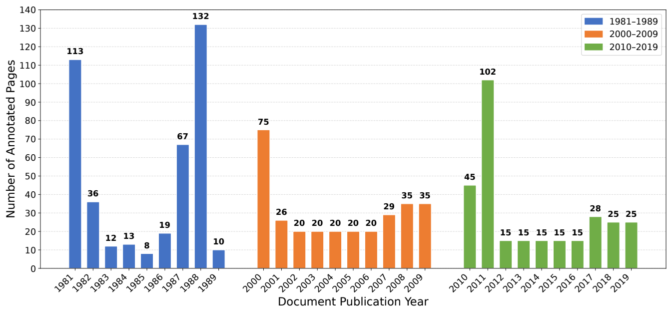

Fig. 2. Distribution of annotated page samples across document publication years (1981–1989 and 2000–2019).

correction, as scans were of poor quality. This gave us 1,010 data pairs: 410 from the 1980s (1981–1989), 300 from the 2000s (2000–2009), and 300 from the 2010s (2010–2019). The decade 1990–1999 was excluded due to the infeasibility of obtaining complete coverage within the manual annotation timeframe. The selected periods maximise the temporal span of the available diachronic periods.

D. Dataset Splits

The 1,010 annotated pairs were randomly shuffled using a fixed random seed of 42 to ensure reproducibility, then partitioned into 707 training, 101 validation, and 202 test pairs following a 70/10/20 ratio. The resulting split maintains balanced representation across all three document eras (1981–1989, 2000–2009, and 2010–2019) in each subset, as documents within each era share similar printing styles and scan characteristics. The final dataset was uploaded to the Hugging Face Dataset Hub as a public repository for version control and reproducibility. Table I summarises the dataset statistics.

IV. EXPERIMENTAL SETUP

A. Model Selection

Three VLMs were selected on the basis of being purpose-built for dense document OCR, script-agnostic recognition, and compatibility with parameter-efficient fine-tuning on available GPU hardware.

DeepSeek-OCR V1 [12] proposes the paradigm of context optical compression. The encoder consists of an 80M parameter SAM-base backbone augmented with a window attention and a 300M parameter CLIP-large backbone with 16× convolutional compression, followed by a DeepSeek-3B

TABLE II
SUMMARY OF ALL 8 FINE-TUNING EXPERIMENTS.

Exp	Model	GPU	Quant.	r	α	Dropout
1	DeepSeek-OCR V1	P100 16 GB	4-bit NF4	16	16	0
2	DeepSeek-OCR V2	A100 80 GB	16-bit	32	64	0.1
3	DeepSeek-OCR V2	A100 80 GB	16-bit	16	16	0
4	DeepSeek-OCR V2	RTX 5090	4-bit NF4	16	16	0
5	LightOnOCR-2-1B	P100 16 GB	4-bit NF4	16	16	0.05
6	LightOnOCR-2-1B	P100 16 GB	4-bit NF4	32	64	0.1
7	LightOnOCR-2-1B	RTX 4090	4-bit NF4	32	64	0.1
8	DeepSeek-OCR V2	RTX 3090	4-bit NF4	16	16	0

MoE decoder that uses 570M parameters at each forward propagation step. All experiments used *Gundam* mode (base size 1024 px, crop size 640 px) for multi-scale document processing. DeepSeek-OCR V1 explicitly lists Sinhala among its 100 supported languages.

DeepSeek-OCR V2 [13] proposes the DeepEncoder V2, which replaces the CLIP model with a Qwen2-0.5B LLM-style architecture and introduces causal flow queries that semantically reorder visual tokens before decoding. The local crop size is upgraded from 640 px to 768 px in Gundam mode, enabling finer character discrimination.

LightOnOCR-2-1B [14] is a compact 1.005B-parameter fully-differentiable VLM consisting of a native-resolution ViT encoder pre-trained with Pixtral, a 4× downsampling multimodal projector, and a Qwen3 language model decoder. Its significantly lower parameter count makes it well-suited to constrained GPU environments, while still supporting complex document layouts, including tables and mathematical formulae.

B. Fine-Tuning with LoRa and QLoRA

All models were finetuned using LoRa and QLoRA [8], [15] through Unsloth [33] for DeepSeek models and through LightOn [34] for LightOnOCR-2-1B. QLoRA maintains the 4-bit quantised NF4 parameters of the base model while adding trainable low-rank adaptation matrices to the attention and feed-forward projections of the transformer decoder. Gradient checkpoint was enabled for all experiments to reduce peak VRAM usage. The table II summarises all 8 experiments, which varied GPU hardware, quantisation level, LoRA rank, and input resolution across the three model families, using the same 707/101/202 data split throughout.

C. Evaluation Metrics

All 8 fine-tuned models and baselines were evaluated on the same 202-sample held-out test set using five metrics: **CER** (Character Error Rate) and **WER** (Word Error Rate), computed via edit distance at the character and word levels respectively (lower is better); and **BLEU**, **METEOR**, and **ANLS** (Average Normalised Levenshtein Similarity), measuring n-gram precision, fluency, and string similarity respectively (higher is better). CER is the primary metric for script-level OCR quality.

V. RESULTS

A. Pre-Trained Baseline Performance

Before fine-tuning, the three VLMs were evaluated on the test dataset (202 samples) in a zero-shot manner. None of the

TABLE III
PRE-TRAINED ZERO-SHOT PERFORMANCE BEFORE FINE-TUNING

Exp	Model	CER↓	WER↓	METEOR↑	BLEU↑	ANLS↑
1	DeepSeek-OCR V1	0.6146	0.7817	0.3754	0.3749	0.3854
2	DeepSeek-OCR V2	0.9611	0.9930	0.1940	0.0585	0.0389
3	LightOnOCR-2-1B	0.8805	0.9896	0.1311	0.0316	0.1195

TABLE IV
FINE-TUNED MODEL PERFORMANCE AFTER QLORA ADAPTATION (202 TEST SAMPLES).

Exp	Model (Config)	CER↓	WER↓	METEOR↑	BLEU↑	ANLS↑
1	DeepSeek-OCR V1	0.0302	0.1227	0.8441	0.9531	0.9698
2	DeepSeek-OCR V2	0.3471	0.4745	0.5191	0.7288	0.6529
3	DeepSeek-OCR V2	0.0694	0.1723	0.7739	0.9153	0.9306
4	DeepSeek-OCR V2	0.0694	0.1828	0.7794	0.9181	0.9306
5	LightOnOCR-2-1B	0.2517	0.3831	0.5342	0.7329	0.7483
6	LightOnOCR-2-1B	0.1413	0.2118	0.6541	0.8332	0.8587
7	LightOnOCR-2-1B (1540px)	**0.0105**	**0.0563**	**0.9492**	**0.9808**	**0.9895**
8	DeepSeek-OCR V2 (r16, RTX3090)	0.0831	0.1975	0.7602	0.9046	0.9169

Bold = best overall.

models performed satisfactorily; DeepSeek-OCR V1 generated a CER of 61.46%. However, DeepSeek-OCR V2 (96.11%) and LightOnOCR-2-1B (88.05%) failed almost completely, which shows that zero-shot Sinhala OCR using present-day VLMs is not possible with current VLMs due to insufficient pre-training coverage [2], [4]. Table III reports the full zero-shot metrics.

B. Fine-Tuning Results

After fine-tuning with QLoRA across 8 experiments, all models showed dramatic improvements over their baselines. Table IV summarises the performance of all 8 fine-tuned models on the same 202-sample test set.

In Experiment 7, we demonstrated LightOnOCR-2-1B[2], which achieved the lowest CER (1.05%) and WER (5.63%). The reason for the success of the longest edge of 1540 px compared to 700 px was the Input resolution, which reduced CER from 25.17% to 1.05% and enabled the model to identify finer Sinhala stroke diacritics. Compared to other runs by DeepSeek-OCR V2[3], Experiment 2 with r = 32 and dropout of 0.1 yielded far lower Compared to Experiments 3 and 4, with r=16 and no dropout, it is confirmed that aggressive regularization is counterproductive on small datasets [26].

C. Benchmarking Against Existing OCR Engines

The three best-fine tuned models (Experiments 7, 1, and 3) were compared against four existing OCR Engines in the same 202-sample real-world test set: Google Document AI, Surya-OCR [2], Tesseract v5 [4], and Subasa-OCR, an open-source Sinhala-specific engine. Results are shown in Table V.

LightOnOCR-2-1B (Exp 7) performed the best with CER of 1.05% and WER of 5.63%, surpassing Google Document AI (CER 2.06%), a paid commercial system without available model weights. Surya-OCR served as the optimal open-source baseline, scoring CER 8.84% and WER 26.64%, although WER was approximately 25 points lower than the stated 2.61%. The score obtained for synthetic Sinhala shows a significant difference between synthetic baselines [2] and real-world degraded documents.

[2]https://huggingface.co/avishadilhara/sinhala-lightonocr-2-1b-Qlora
[3]https://huggingface.co/avishadilhara/sinhala-deepseek-ocr-Qlora

TABLE V
BENCHMARKING FINE-TUNED MODELS VS. EXISTING OCR ENGINES ON THE `SINHALA-OCR-LK-ACTS-1010` TEST SET (202 SAMPLES).

Rank	Model	CER↓	WER↓	METEOR↑	BLEU↑	ANLS↑
1	LightOnOCR-2-1B (Exp 7)	**0.0105**	**0.0563**	0.9492	**0.9808**	**0.9895**
2	Google Document AI	0.0206	0.0826	**0.9609**	0.9724	0.9797
3	DeepSeek-OCR V1 (Exp 1)	0.0302	0.1227	0.8441	0.9531	0.9698
4	DeepSeek-OCR V2 (Exp 3)	0.0694	0.1723	0.7739	0.9153	0.9306
5	Surya-OCR	0.0884	0.2664	0.7806	0.8877	0.9116
6	Subasa-OCR	0.0999	0.5251	0.7483	0.8355	0.9006
7	Tesseract v5	0.1069	0.4514	0.7947	0.8117	0.8951

Bold = best overall.

TABLE VI
PERIOD-LEVEL WEIGHTED MEAN CER (%) AND WER (%) FOR FINE-TUNED MODELS AND BASELINES.

	1981–89 (High)		2000–09 (Mod.)		2010–19 (Modern)	
Model	CER	WER	CER	WER	CER	WER
LightOnOCR-2-1B (Exp 7)	**1.66**	**8.70**	**0.71**	4.42	**0.58**	2.81
DeepSeek-OCR V1 (Exp 1)	4.65	17.62	2.35	9.64	1.54	7.78
Google Document AI	3.26	15.86	1.30	**4.36**	1.18	**2.01**
Surya-OCR	12.80	37.40	7.35	22.24	5.15	16.85
Tesseract v5	18.07	69.86	7.62	32.64	4.05	24.66
Subasa-OCR	14.82	67.19	8.07	46.01	5.54	39.60

Bold = best per period per metric.

VI. DIACHRONIC ANALYSIS

A. Temporal Distribution of the Test Set

The test set spans three print periods: **1981–1989** (n=82, high degradation ink fading, bleed-through, and noisy); **2000–2009** (n=54, moderate degradation, desktop publishing era with minor defects); and **2010–2019** (n=66, modern digitally produced with minimal degradation) [10], [16], [17].

B. Period-Level Aggregated Performance

Table VI reports the sample-weighted mean CER and WER for all six models across three periods. Period-level scores are computed as sample-weighted means across individual publication years to account for unequal representation per year.

In all cases, there is a clear diachronic decline pattern in CER and WER, with values increasing as documents become older and more physically degraded (Figs. 3 and 4). This result confirms the hypothesis proposed by [16], [17], which posits that OCR accuracy deteriorates systematically with historical print quality, and provides an innovative application of this theory to the Sinhala language for the first time.

C. Legacy Baselines: Diachronic Collapse

The findings from the diachronic analysis highlight a key shortcoming of traditional OCR tools in handling realistic historical Sinhala documents. In this regard, the OCR tools Tesseract v5 and Subasa-OCR achieve WERs of 69.86% and 67.19% respectively on the 1980s documents, suggesting that around seven out of ten words are misrecognized. Therefore, line-segmentation-based OCR systems, which are built for working with clean binary images, are essentially incapable of handling degraded historical Sinhala documents [30]. Although Surya-OCR is currently the best-performing open-source baseline with a WER of 37.40% in the 1980s, its performance is significantly inferior compared to the reported WER of 2.61% on simulated documents [2] a gap, confirming that synthetic benchmarks substantially overestimate real-world performance.

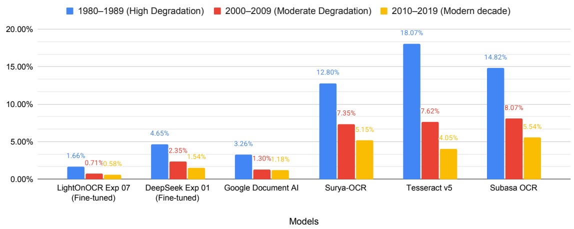

Fig. 3. Sample-weighted mean CER across three temporal periods for all six models.

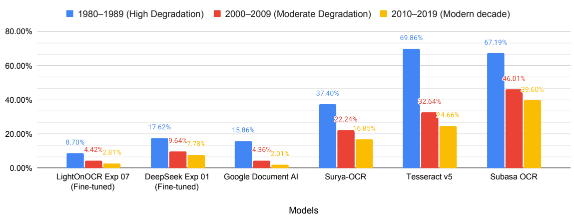

Fig. 4. Sample-weighted mean WER across three temporal periods

D. BLEU, METEOR, and ANLS Across Periods

Table VII reports BLEU, METEOR, and ANLS across the three periods, confirming that the trends observed in CER/WER are consistent across all five evaluation metrics. LightOnOCR-2-1B achieves BLEU scores of 97.15%, 98.50%, and 98.90% across the three periods, indicating that high n-gram precision is maintained even on degraded 1980s pages.

VII. Discussion

A. Fine-Tuning, and Commercial Comparison

With only 707 training data points in the real world, QLoRA-based fine-tuning enabled all three VLMs to move from being unable to recognise Sinhala words to becoming highly accurate. LightOnOCR-2-1B lowered its CER from 88.05% to 1.05% (with an overall improvement of 86.97%), whereas DeepSeek-OCR V1 went from 61.46% to 3.02%, following the same pattern as other work showing that PEFT based on LoRA requires far fewer samples than full fine-tuning for low-resource Indic scripts [1], [9].

Fine-tuned LightOnOCR-2-1B (CER 1.05%) outperformed Google Document AI (CER 2.06%) according to the main measure, but Document AI had a slight advantage on METEOR (96.09% compared to 94.92%). This analysis is clear: Document AI output was solely used as an initial annotation seed and manually corrected character by character before being used as ground truth, avoiding circularity in the evaluation.

B. Diachronic Degradation and Dataset Significance

Diachronic degradation is a common indicator for all six systems: LightOnOCR-2-1B CER increases by 2.9× from the contemporary period (0.58%) to the highly degraded period of 1981–1989 (1.66%), whereas Tesseract v5 increases by 4.5× (4.05% → 18.07%). Subasa-ocr attains WER 67.19% on texts from the 1980s, demonstrating that almost seven out of ten words are recognized incorrectly [16]. Notably, the reduced diachronic sensitivity of these fine-tuned VLMs can be attributed to their domain-specific training using actual degraded data, and not any inherent structural superiority over line segmentation models [4], [26]. The performance of a fine-tuned line-level model in achieving similar resilience remains an open question.

TABLE VII

Period-level weighted mean BLEU, METEOR, and ANLS for fine-tuned models and baselines.

	1981–89			2000–09			2010–19		
Model	BLEU	MTR	ANLS	BLEU	MTR	ANLS	BLEU	MTR	ANLS
LightOnOCR (Exp 7)	**97.15**	91.11	**98.34**	**98.50**	97.52	**99.29**	**98.90**	97.54	**99.42**
DeepSeek V1 (Exp 1)	93.17	75.97	95.35	96.13	89.27	97.65	97.31	90.92	98.46
Google Doc AI	95.65	**92.25**	96.79	98.17	**97.91**	98.71	98.47	**99.38**	98.84
Surya-OCR	84.01	66.14	87.20	90.60	82.37	92.65	93.18	89.34	94.85
Tesseract v5	70.01	58.99	83.30	85.98	87.66	90.93	91.10	98.21	96.07
Subasa-OCR	74.87	57.35	85.26	87.56	81.43	91.94	91.05	91.16	94.47

MTR = METEOR. All values in %.
Bold = best per period per metric.

C. Limitations

There Several limitations constrain the validity of these findings and should be addressed in future work.

Domain homogeneity - The corpus contains only Sri Lankan legislative statutes. Although this guarantees consistency, it limits the diversity of fonts, layout, and vocabulary. The performance could vary widely across Sinhala in newspapers, textbooks, and handwritten Sinhala.

Temporal coverage gap - The decade from 1990 to 1999 was excluded due to the infeasibility of manual annotation within the project timeframe.

Year-level sample imbalance - There is a large variation samples for per year within each time period; for example, 1988 has 26 pages available, whereas some years have only 1 to 2 pages available. Time-period-level weighted averages solve this problem, yet year-based point averages remain statistically unreliable.

VIII. Conclusion

The fine-tuned LightOnOCR-2-1B (Exp 7) model obtained a CER of **1.05%** and a WER of **5.63%** on the realistic test set, outperforming all the open-source baselines and Google Document AI (CER 2.06%) models, proving that a small 1B-parameter open-source VLM optimized on 707 realistic samples can outperform a commercial OCR engine in this setting. For the first time, a diachronic evaluation of Sinhala OCR was conducted, showing that the degradation of documents during printing periods is a universal factor influencing performance in all six models, increasing the CER by a factor of 2.9× in the case of LightOnOCR-2-1B and 4.5× in the case of Tesseract v5 when comparing the modern and highly degraded periods. Importantly, the decreased sensitivity of the optimized models to diachronic degradation cannot be explained by architectural differences but rather by domain-specific optimization on degraded examples, which remains a question for future experimental studies.

Further research can include expanding the corpus to include additional types of Sinhala documents (newspapers, textbooks, gazettes), the missing decade 1990-1999, and conducting a diachronic comparison using controlled architectural experiments with fine-tuned line-level and page-level models.

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